

So, What is Project Pulse?

Our Senior design project focuses on the modification of a Mark II TRIGA reactor. Specific enhancements include; enlarging the central cavity, increasing the number of transient rods, and increasing the number of fuel rods. In this, achieving the goals of being able to obtain a larger peak power and having a more versatile central cavity compared to TRIGA class reactors.

What is a TRIGA Mark II reactor?

The TRIGA Mark II is a type of research nuclear reactor designed for safety, reliability, and flexibility. It uses uranium fuel embedded in a zirconium hydride matrix, which gives it a unique negative temperature coefficient. Meaning that the reactor naturally with the increase in temperature becomes less reactive. This feature makes TRIGA reactors inherently safe, especially during pulsing operations.

Where did we get the inspiration?

Group interest in pulsed reactors since these types of reactors are often used for national security applications and there is a lack of pulsed power topics in the undergraduate curriculum. These two reasons sparked the passion for this senior design project.

Who would be our potential buyers?

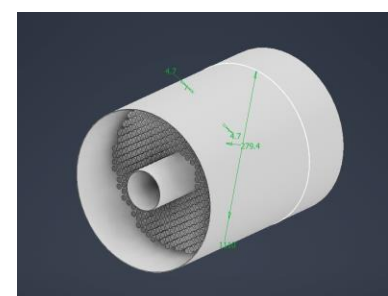
Since our type of reactor is often used to simulate weapons effects on different types of materials and electronics, a potential contract with the DoD or with National Laboratories would be ideal. However, the versatility of our reactor universities could benefit from training students on reactor operations and pertinent research.

Design Criteria

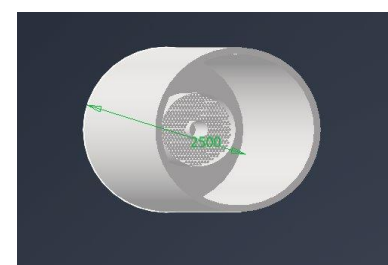
When it came to modifying the Mark II TRIGA, there needed to be adjustments made to the dimensions of the reactor's system. Our groups updated dimensions can be seen below in the table

Core Dimensions		
What?	Value	Units
Fuel Rod Length	73.152	cm
Fuel Rod Inner diameter	3.5814	cm
Fuel rod Outer diameter	3.7338	cm
Cladding Thickness	0.0762	cm
Core Diameter	111	cm
Core Length	150	cm
Rabbit hole	27.94	cm

The updated dimensions enabled us to create a CAD model to better visualize the core layout and modifications to original system. These CAD Models can be seen below.



This is the entire core for our modified system. In this model is the total amount of rods and the core barrel.

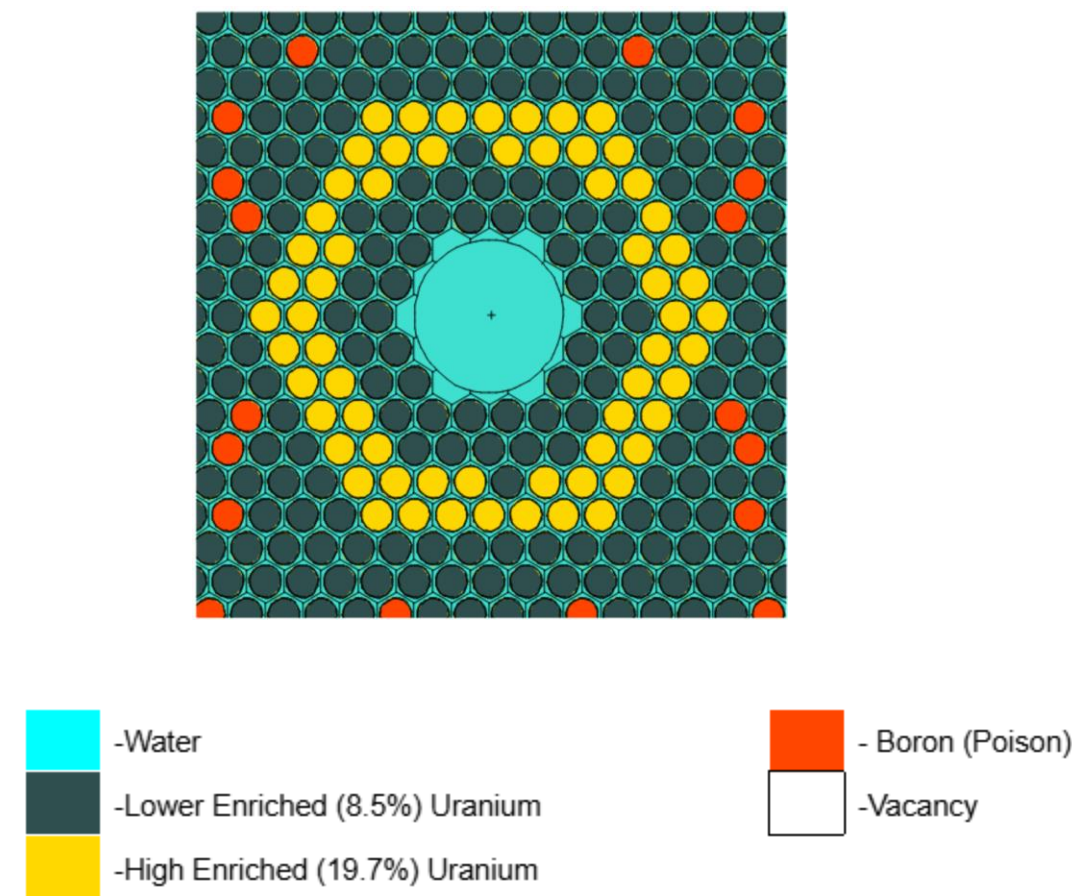
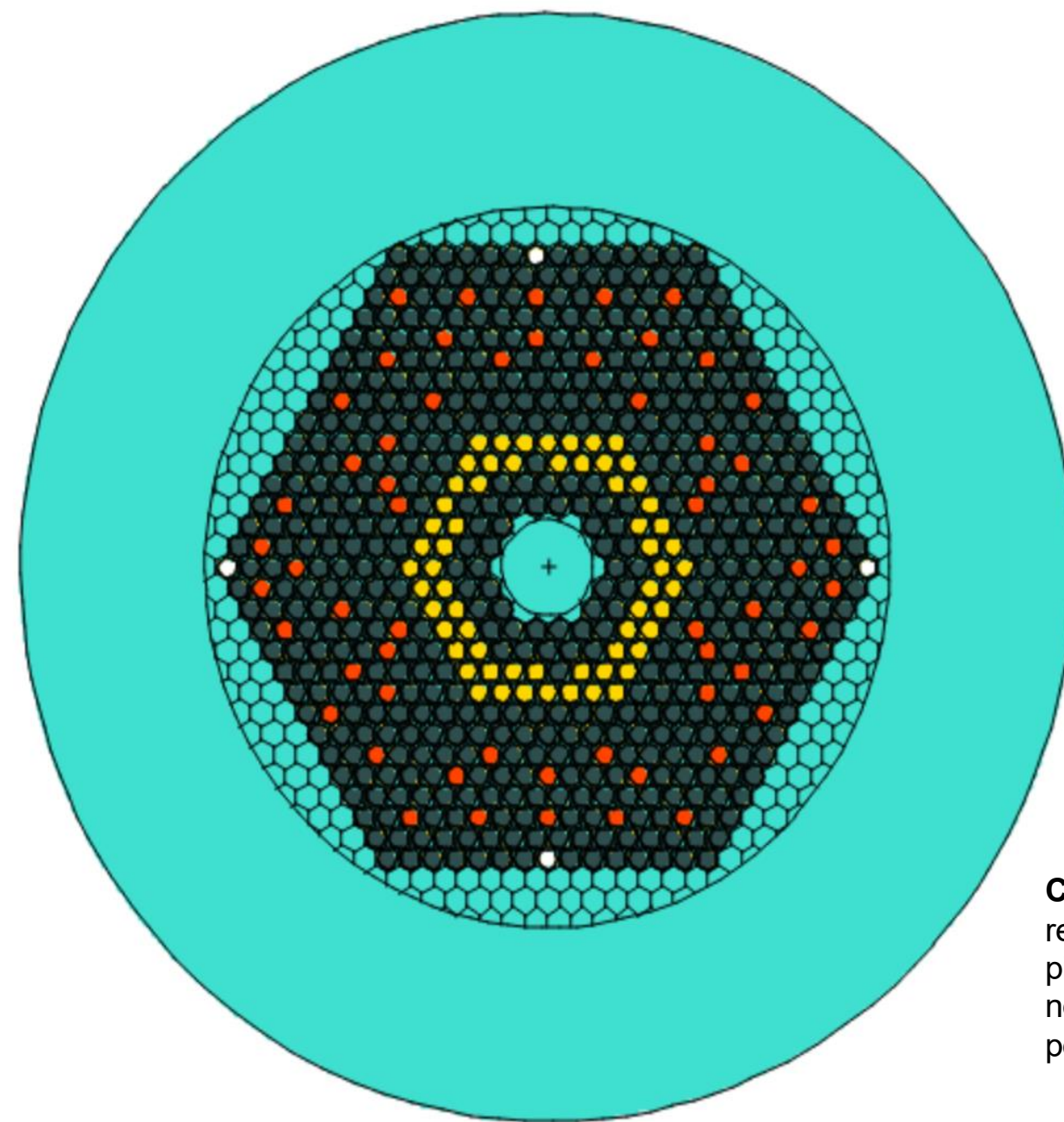


This is the core barrel with the core inside of the reactor bucket.

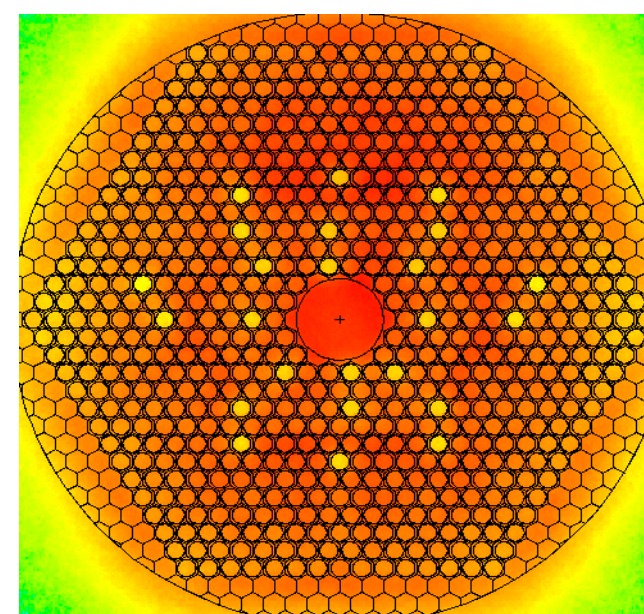
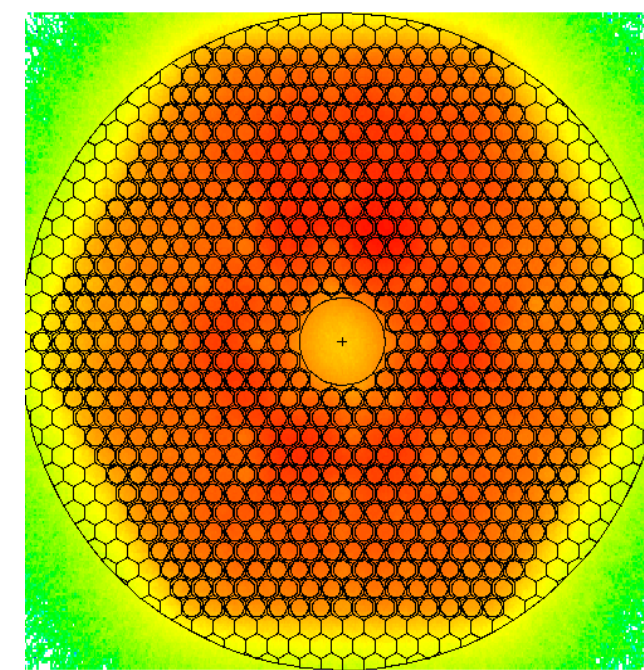
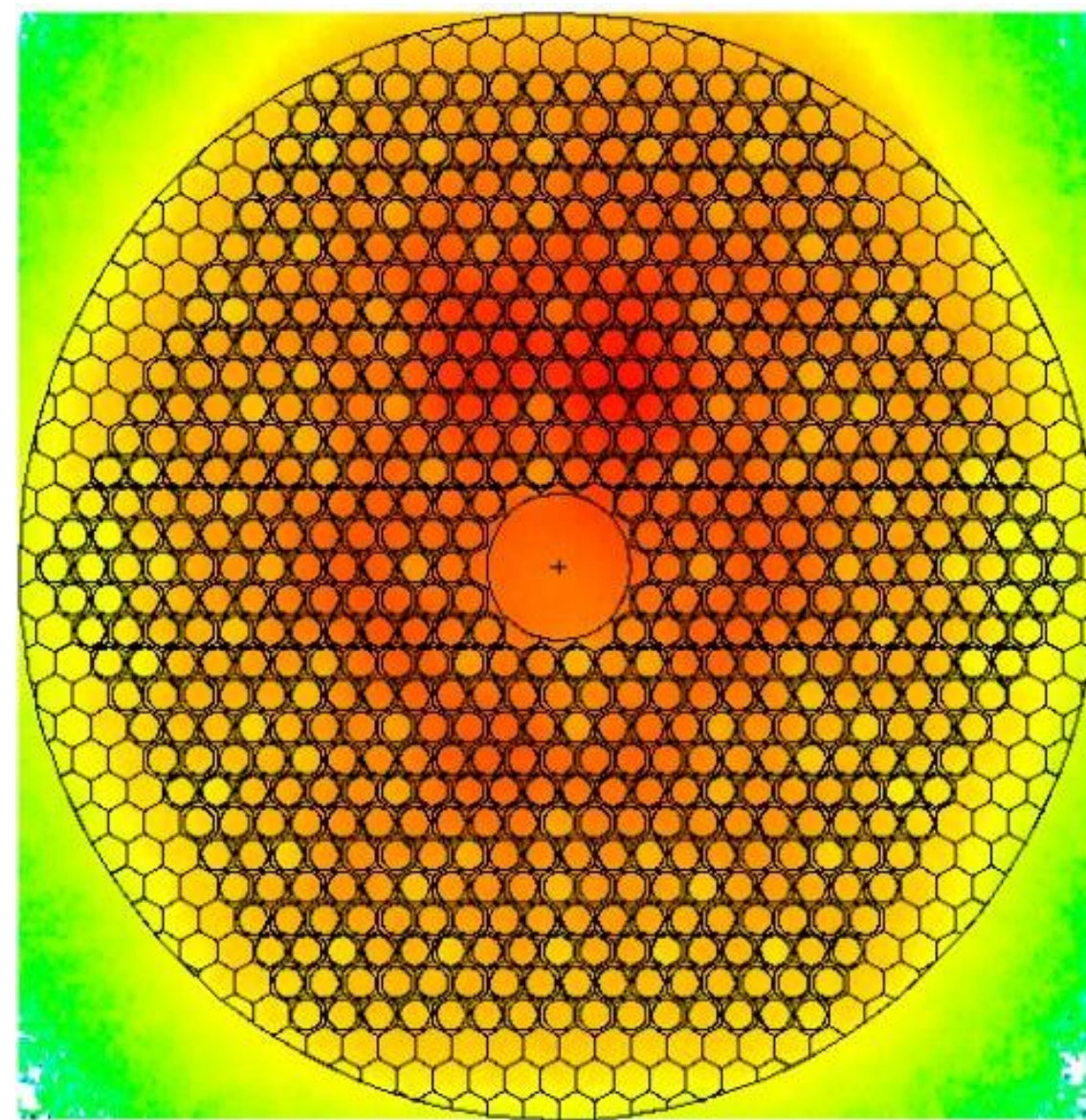


This is a different view to note the location of the core barrel inside the reactor bucket.

Core Neutronics

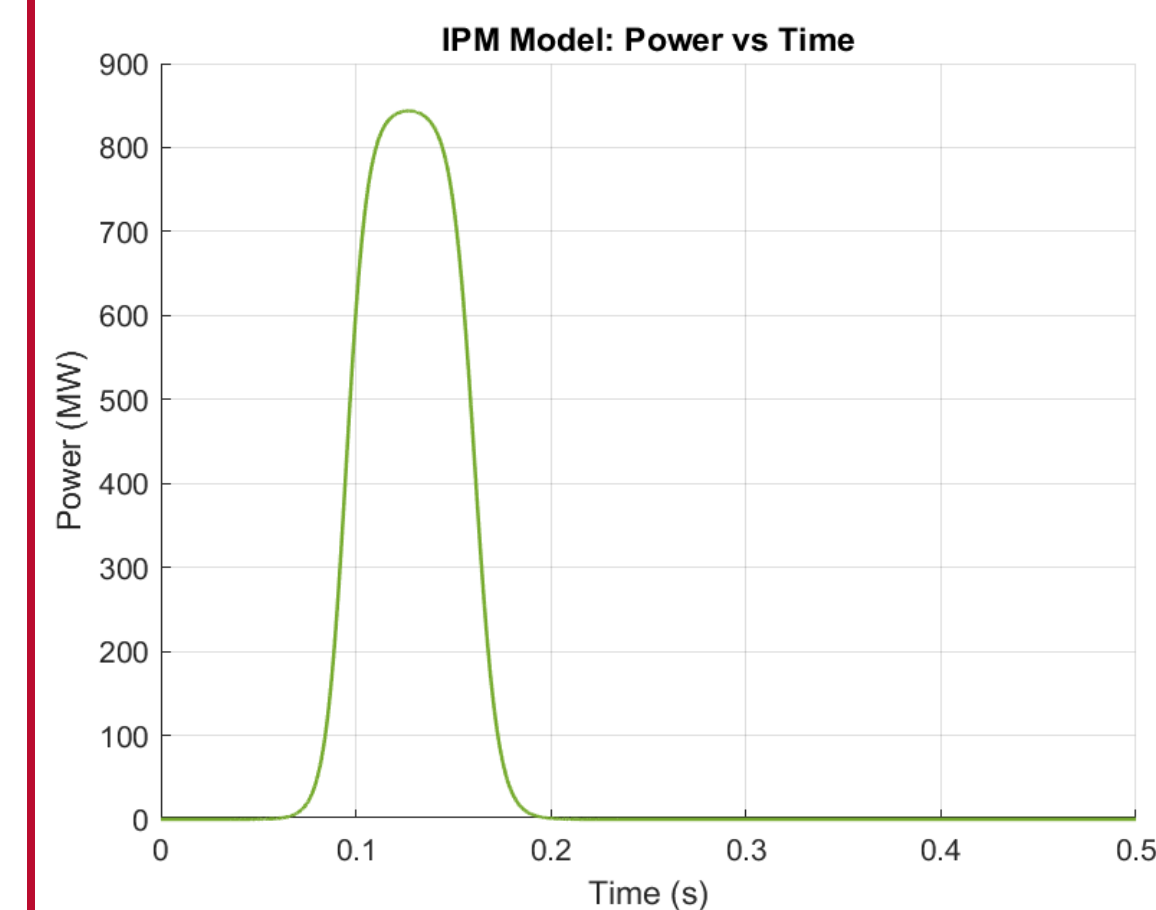
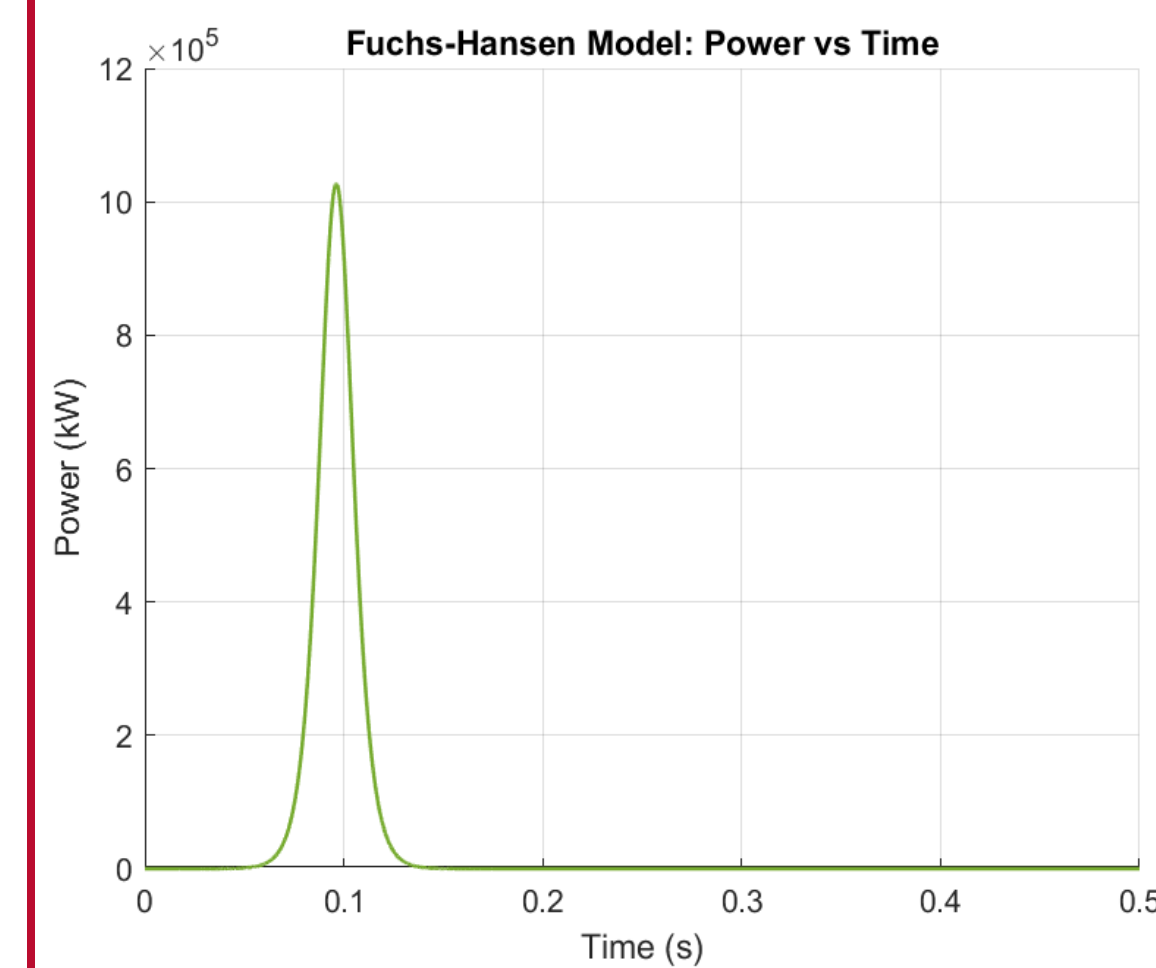
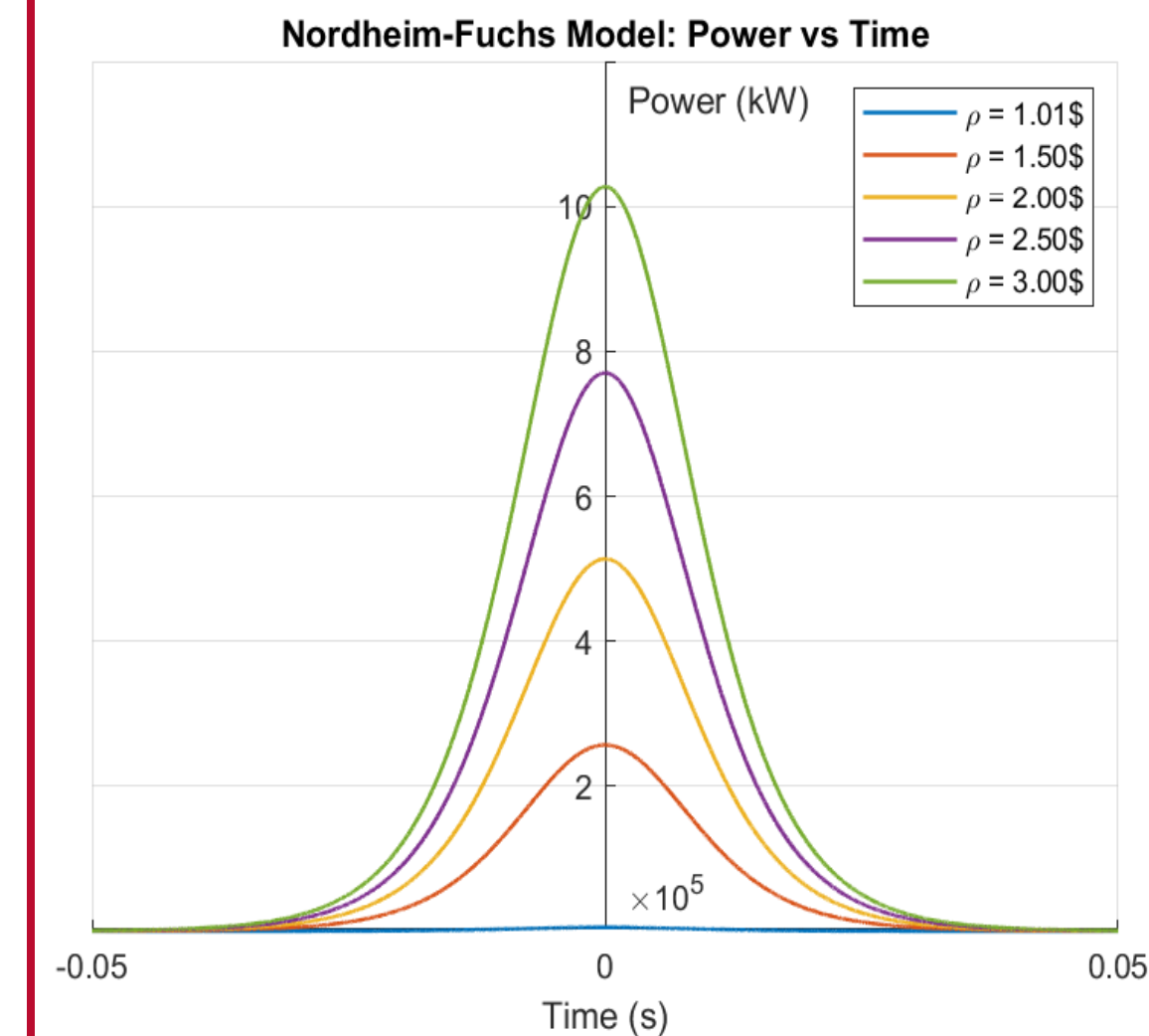


Core design: The core is made of a lattice of hexagonal cells; each cell representing a rod, and each color representing a material. The above picture highlights the core's main cavity, where samples will be exposed to neutron flux for experiments. Core images are from top-down viewing position.



Neutron Flux: One of the core design goals of any pulse reactor is a quick burst of extremely high neutron flux. The figure above shows how this flux is distributed in the core. At its most reactive, with all transient rods pulled out, the reactor reaches a multiplication factor of 1.13. To the right are similar flux profiles on the core based on energy level, with the top right being fast flux and bottom right being thermal flux. On the thermal flux the transient rods are down and can be seen as the cells with low flux.

Pulse Transients



Transient Calculations

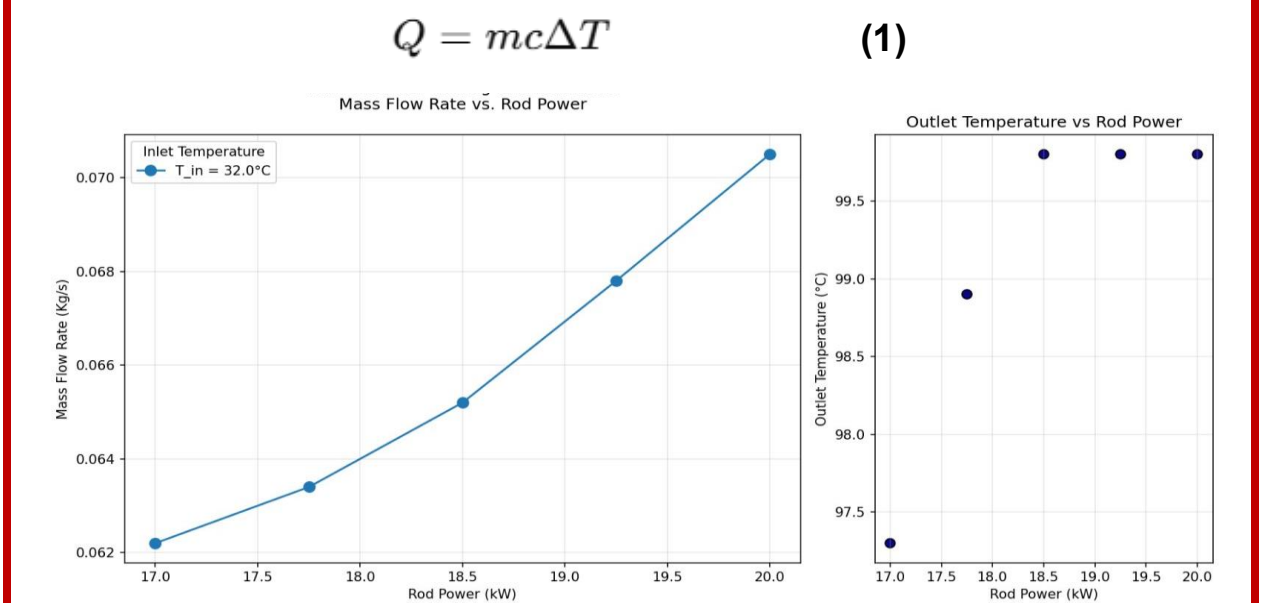
Calculations of power transient models were done using MATLAB. The Improved Pulse Model (IPM) is the most accurate model of our pulse. This model considers the effects due to delayed neutrons unlike the other two models. Also, this model uses Newtonian Laws of Cooling to derive how much power is put into the system. However, more iterations are needed for the IPM model to accurately reflect our reactor core parameters.

Pulse Model	Max Pulse Power (MW)	Time of Max Power (sec)
Nordheim-Fuchs	1,027	N/A
Fuchs-Hansen	1,027	0.0962
IPM	844	0.1273
Reference ITU TRIGA MK II	1,200	0.0200
Reference Thai Research Reactor	2,000	0.0105

Used reactor period of known TRIGA to evaluate all relevant power transients. MCNPX model would be needed to obtain actual modeled reactor period.

Thermal Hydraulics

Thermal Hydraulics calculations performed looked at the relationship of the mass flow rate through the core when it was operating at steady state power. A model was developed using Equation (1) as the foundation with variables for a triangular pitch channel.



Mass Flow Rate: The code developed does the following given our reactor parameters;

- Calculates natural circulation mass flow rate by balancing buoyancy with pressure losses (friction, form, acceleration).
- Considers temperature-dependent fluid properties: density, viscosity, and specific heat.
- Uses channel geometry to determine flow area and hydraulic diameter.
- Iteratively refines outlet temperature and Reynolds number for accurate heat transfer modeling.

Future Work

- Neutronic effects on local environment.
- Forced Convection Thermal Hydraulics.
- Improved MCNP Model.

References

